

DIGITAL AGRO FIRM

A Low-Cost Solar Powered Smart Agriculture System



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Live: <https://smartfarm.mostakinahmed.com>

Abstract.

Agriculture is a cornerstone of Bangladesh's economy, employing nearly 40% of the workforce. However, traditional farming remains inefficient, often leading to the wastage of water and energy due to manual monitoring and data-poor decision-making. These inaccuracies result in significant financial losses for farmers and environmental strain.

To address these challenges, we propose Digital Agro Farm—a low-cost, solar-powered smart agriculture system. By integrating IoT sensors (soil moisture, temperature, and humidity) with an ESP32 microcontroller, the system automates irrigation based on real-time data. To further optimize resources, a Weather API prevents unnecessary irrigation if rain is forecasted.

Beyond automation, the project includes a motion-based security system for crop protection and a MERN-based dashboard for remote monitoring. By utilizing sustainable solar energy, this project offers an eco-friendly and affordable technological framework to modernize farming, improve efficiency, and support the sustainable development of Bangladesh's agricultural sector.

1. Introduction

Agriculture remains vital, yet traditional methods face critical challenges: inefficient manual irrigation, high energy costs, and crop destruction by animals. In rural areas, the lack of a stable power grid often forces farmers to rely on environmentally harmful diesel engines. Furthermore, a lack of data-driven insights into soil quality and manual labor intensity often leads to low crop yields and significant financial losses.

The Digital Agro Farm project addresses these issues through an automated, low-cost IoT ecosystem. Utilizing an array of sensors—including Soil Moisture, pH, Temperature, Motion, and Raindrop sensors—the system analyzes environmental data to automate irrigation and provide actionable insights. By integrating Weather APIs, the system optimizes water usage based on forecasts, while an Infrared/Motion-based security system protects crops from animal intrusion.

The system is powered by renewable solar energy, making it ideal for off-grid rural farms and greenhouses. With a dedicated MERN-based dashboard and mobile application, farmers can monitor real-time conditions and manage their land efficiently, transitioning from labor-intensive manual methods to high-yield, sustainable precision agriculture.

2. Problem Statement

Agriculture in Bangladesh continues to rely heavily on traditional farming practices that depend on manual monitoring and decision-making. Farmers often irrigate fields without accurate measurement of soil moisture, temperature, or humidity, which leads to excessive water consumption and inefficient use of electricity or diesel-powered pumps. In many rural areas, unreliable electricity supply forces farmers to use diesel engines for irrigation, increasing production costs and causing environmental pollution. The lack of real-time data and technological support reduces farming efficiency and limits crop productivity.

Additionally, poor soil management and insufficient knowledge about soil conditions such as pH levels and nutrient content contribute to declining soil fertility and lower yields. Farmers frequently make decisions based on assumptions rather than precise data, which affects crop growth and overall farm performance. Animal intrusion and unexpected weather changes further increase the risk of crop damage, leading to significant financial losses. The absence of an integrated monitoring and alert system makes it difficult for farmers to respond quickly to these challenges.

Therefore, there is a pressing need for an affordable, eco-friendly, and automated smart agriculture system that can monitor real-time field conditions, optimize irrigation, reduce energy consumption, and protect crops from environmental and external threats. A technology-driven solution that integrates sensors, automation, renewable energy, and weather forecasting can significantly improve farming efficiency, sustainability, and productivity in Bangladesh.

3. Objectives

3.1 To design a low-cost, solar-powered smart agriculture system

The primary objective of this project is to design and implement a budget-friendly smart agriculture system suitable for small and medium-scale farms in Bangladesh. Since many rural areas suffer from unreliable electricity and high fuel costs, the system is powered by solar energy, making it eco-friendly and independent of grid electricity or diesel engines. The use of affordable sensors and commonly available hardware ensures that the system remains accessible to farmers with limited financial resources.

3.2 To automate irrigation using soil moisture, temperature, and humidity sensors

This project aims to automate the irrigation process by continuously monitoring soil moisture, temperature, and humidity levels using sensors. Instead of relying on manual judgment or fixed schedules, the system analyzes real-time data and automatically controls the water pump. This ensures that crops receive the right amount of water at the right time, improving crop health, reducing farmer workload, and increasing overall productivity.

3.3 To reduce water wastage using weather-based decision making

Water is a critical and limited resource in agriculture. One of the key objectives is to minimize unnecessary water usage by integrating real-time weather data through a weather forecast API. If rainfall is predicted within the next few hours, the system delays irrigation automatically. This intelligent decision-making approach helps conserve water, prevents over-irrigation, and supports sustainable farming practices.

3.4 To protect crops using motion-based security sensors

Crop damage caused by animals is a major concern for farmers, often resulting in significant yield loss. This project aims to protect crops by implementing a motion-based security system using sensors such as PIR or infrared sensors and lasers. When movement is detected in the field, the system can trigger alerts or warning mechanisms, helping farmers take timely action and reducing crop loss.

3.5 To suggest proper fertilizer usage based on soil condition

Poor soil management and lack of knowledge about soil quality often lead to low crop yield. Another important objective of this project is to analyze soil conditions (such as moisture level and pH value) and provide fertilizer recommendations accordingly. This helps farmers apply the correct type and amount of fertilizer, improving soil health, reducing excessive chemical use, and increasing long-term agricultural productivity.

3.6 To implement a mobile application for real-time monitoring

The project also aims to develop an optional mobile application that allows farmers to monitor their farms remotely. Through the application, users can view real-time sensor data, receive notifications about irrigation status, weather conditions, and security alerts. This feature enables farmers to make informed decisions anytime and anywhere, improving farm management efficiency and convenience.

4. Proposed System Overview

Digital Agro Farm is an IoT-driven, solar-powered ecosystem designed to modernize traditional farming through automation and data-driven decision-making. At its core, an ESP32 microcontroller serves as the central hub, aggregating real-time data from an array of soil moisture, temperature, rainfall, and ultrasonic sensors. This data is processed locally and transmitted via Wi-Fi to a Node.js backend and stored in MongoDB, providing a historical record for long-term agricultural analysis.

The system achieves intelligent irrigation by controlling a DC water pump through a relay module. Rather than relying on manual timers, irrigation is triggered by precise soil moisture thresholds and refined by the OpenWeatherMap API. If rain is detected or forecasted, the system automatically overrides and postpones scheduled watering to maximize conservation. To eliminate dependence on unstable rural grids or hazardous diesel engines, the hardware is entirely solar-powered. A photovoltaic panel, managed by a charge controller and a rechargeable battery, ensures 24/7 operation, making the system carbon-neutral and cost-effective.

Crop protection is managed by a security module utilizing PIR and laser or motion sensors to detect animal intrusion, triggering local alerts and real-time push notifications. Farmers can monitor the entire farm through a custom React.js web dashboard or a mobile application, enabling remote management from any location. Overall, this scalable solution improves agricultural productivity and reduces resource

wastage by combining professional software architecture with sustainable engineering practices.

5.Feature List

5.1 Solar-Powered System

- Fully powered by renewable solar energy
- Works in off-grid rural areas
- Includes battery backup for 24/7 operation

5.2 Smart Irrigation System

- Automatic irrigation based on real-time soil moisture data
- Controls water pump using relay module
- Prevents overwatering and underwatering

5.3 Weather-Based Irrigation Control

- Integrates Weather API (OpenWeatherMap)
- Delays irrigation if rain is forecasted
- Reduces unnecessary water usage

5.4 Environmental Monitoring

- Monitors:
 - Soil moisture
 - Temperature
 - Humidity
 - Rainfall
- Provides accurate real-time field conditions

5.5 Soil Analysis & Fertilizer Suggestion

- Analyzes soil condition (moisture + pH)
- Suggests appropriate fertilizer usage
- Improves crop yield and soil health

5.6 Crop Security System

- PIR motion sensor and laser to detect animal intrusion
- Buzzer alert for immediate warning
- Real-time alerts sent to user

5.7 Fire Detection System

- Flame sensor detects fire hazards
- Triggers alerts for safety
- Helps prevent crop and property damage by spraying water

5.8 Water Level Monitoring

- Monitors irrigation tank/reservoir level
- Prevents dry running of pump
- Ensures efficient water management

5.9 IoT Connectivity (ESP32-Based)

- Uses ESP32 with built-in Wi-Fi
- Sends sensor data to cloud server
- Enables remote control and monitoring

5.10 Web Dashboard

- Real-time visualization of sensor data
- Displays:
 - Soil moisture trends
 - Temperature & humidity

- Irrigation status
- Built using [React.js](#)

5.11 Mobile Application Support

- Monitor farm remotely via smartphone
- Receive notifications:
 - Irrigation status
 - Weather alerts
 - Security alerts
- Control irrigation manually

5.12 Real-Time Alert System

- Instant notifications for:
 - Low water level
 - Motion detection
 - Fire detection
- Push notifications via app

5.13 Automated Decision System

- Uses predefined thresholds
- Makes smart decisions without human intervention
- Reduces manual labor and errors
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5.14 Cloud Data Storage & Analytics

- Stores historical data in MongoDB
- Enables trend analysis
- Helps in future farm planning

5.15 Remote Control Capability

- Users can:
 - Turn irrigation ON/OFF remotely
 - Monitor system status from anywhere

6. Methodology

The methodology of the Digital Agro Farm project describes the systematic approach followed to design, develop, and implement a solar-powered smart agriculture system. The system integrates sensor-based data acquisition, automated decision-making, cloud connectivity, and real-time monitoring to improve farming efficiency while reducing water and energy wastage.

6.1 System Design and Planning

The first step of the methodology involves analyzing the problems faced by traditional farming systems, such as inefficient irrigation, excessive water usage, lack of soil monitoring, crop damage by animals, and unreliable power supply. Based on these challenges, system requirements were identified.

A modular architecture was designed, dividing the system into the following subsystems:

- Power management subsystem
- Sensor data acquisition subsystem
- Control and automation subsystem
- Security subsystem
- Cloud and application subsystem
- Fire detection and protection system

This modular approach ensures scalability, maintainability, and cost-effectiveness.

6.2 Hardware Setup and Integration

The ESP32 microcontroller is used as the central processing unit. All sensors and actuators are interfaced with the ESP32 using appropriate GPIO pins.

Steps involved:

1. Capacitive soil moisture, temperature, humidity, raindrop, water level, PIR, laser and ultrasonic sensors are installed at appropriate locations in the farm.
2. The DC water pump, relay module, servo motor, and buzzer are connected as output devices.
3. A solar panel, rechargeable battery, and charge controller are integrated to form an independent power supply system.
4. Proper voltage regulation and isolation are ensured to protect sensitive components.

This setup enables continuous environmental monitoring and automation.

6.3 Sensor Data Acquisition

Each sensor continuously collects environmental and soil-related data:

- Soil moisture sensor measures soil water content.
- DHT22 provides ambient temperature and humidity.
- Raindrop sensor detects rainfall.
- A water level sensor monitors the irrigation reservoir.
- PIR, laser and ultrasonic sensors detect motion and obstacles.
- Flame sensor

The ESP32 reads sensor values at predefined intervals and converts raw signals into meaningful data.

6.4 Data Processing and Decision-Making

The ESP32 processes the collected sensor data using embedded C++ logic. Threshold values are predefined based on crop requirements.

Irrigation Logic:

- If soil moisture falls below the threshold and no rain is detected, irrigation is activated.
- If rain is detected or rainfall is forecasted, irrigation is paused.
- If the water level is low, the pump is disabled to prevent damage.

Security Logic:

- Motion detection triggers alerts and activates the buzzer.
- Detects objects with the help of laser and LDR and activates the buzzer.
- Distance detection assists in gate or area monitoring.
- Fire detection and protection system

This automated decision-making minimizes human intervention and errors.

6.5 Weather Api Integration

To enhance irrigation accuracy, weather data is fetched from the OpenWeatherMap API through the backend server.

- The backend periodically retrieves weather forecasts.
- Rain probability and humidity data are analyzed.
- The irrigation schedule is dynamically adjusted based on predicted rainfall.

This approach significantly reduces water wastage.

6.6 Cloud Communication and Data Storage

The ESP32 transmits sensor data to the backend server using HTTP/HTTPS REST APIs.

The backend, developed using Node.js and Express, performs:

- Data validation and processing
- Storage of sensor logs in MongoDB
- Alert generation for abnormal conditions

Historical data enables trend analysis and future optimization.

6.7 Real-Time Monitoring and Control

A web dashboard developed using React.js displays real-time sensor readings, irrigation status, and security alerts.

- WebSockets (Socket.io) enables live updates.
- Users can remotely turn irrigation ON/OFF.
- System notifications are displayed instantly.

This ensures effective farm monitoring from any location.

6.8 Mobile Application Integration

An optional mobile application mirrors the web dashboard functionality:

- Displays live farm data
- Sends push notifications for critical events
- Allows quick decision-making on the go

This improves accessibility for farmers with smartphones.

6.9 Security and Alert Mechanism

The system continuously monitors motion and intrusion events:

- PIR sensors detect animal movement.
- Buzzer alerts locally.
- Notifications are sent to the user via the application.
- This reduces crop damage and economic loss.

6.10 Testing and Validation

The system undergoes multiple testing phases:

- Sensor accuracy testing
- Irrigation automation testing
- Power system testing under different weather conditions
- Network and communication reliability testing

Performance is evaluated based on water savings, response time, and system stability.

6.11 Deployment and Maintenance

The system is deployed in a real or simulated farm environment. Regular calibration of sensors and periodic software updates ensure long-term reliability and scalability.

6.12 Methodology Flow Summary

1. Data collection from sensors
2. Local processing on ESP32
3. Weather data integration
4. Automated decision execution
5. Cloud storage and analysis
6. Real-time monitoring and alerts

7. Hardware Components

7.1 Control & Connectivity

- **ESP32 Microcontroller:** The "brain" of the system, providing dual-core processing and built-in Wi-Fi/Bluetooth for IoT capabilities.

7.2 Power Management System

- **Solar Panel:** High-efficiency photovoltaic cell for renewable energy harvesting.
- **Rechargeable Battery:** Li-ion or Lead-Acid storage for 24/7 operation.
- **Solar Charge Controller:** To regulate voltage and protect the battery from overcharging.

7.3 Environmental & Soil Monitoring

- **Capacitive Soil Moisture Sensor:** To monitor underground water levels without corrosion issues.
- **DHT22 Temperature & Humidity Sensor:** Provides high-accuracy ambient weather data.
- **Raindrop Sensor:** Detects rainfall to automatically override and pause the irrigation schedule.
- **Water Level Sensor:** Monitors the reservoir or tank level to prevent the pump from running dry.

7.4 Security & Automation

- **PIR Motion Sensor:** Detects unauthorized movement or animal intrusion within the farm perimeter.
- **Ultrasonic Sensor (HC-SR04):** Used for distance measurement and object detection near gates or storage areas.
- **Laser and LDR:** Detects unauthorized movement or animal intrusion by using a laser beam directed at an LDR sensor; when the beam is interrupted, the change in light intensity signals an intrusion within the farm perimeter.
- **SG90 Servo Motor:** For automated gate control or adjusting the orientation of the solar panel/sensors.
- **Flame Sensor:** For fire detection.

7.5 Actuators & Indicators

- **Relay Module (5V/12V):** Acts as an electronic switch to control high-power devices.
- **DC Water Pump:** Submersible pump for the smart irrigation system.
- **Active Buzzer:** For local audible alerts during security breaches or low-water emergencies.

8. Software and Tools

8.1 Custom Full-Stack IoT Platform

- **React.js (Frontend):** A custom-built web dashboard for real-time data visualization. It uses **Tailwind CSS** for a responsive UI and **Chart.js** to display live sensor trends (soil moisture, temperature, etc.)
- **Node.js & Express (Backend):** The core API server that handles data ingestion from the ESP32 and serves it to the frontend via RESTful endpoints or WebSockets.
- **MongoDB (Database):** A NoSQL database used to store historical sensor logs, system alerts, and user authentication data.
- **Host-Sheba:** Used for hosting the web application to ensure global accessibility.

8.2 Development Environment

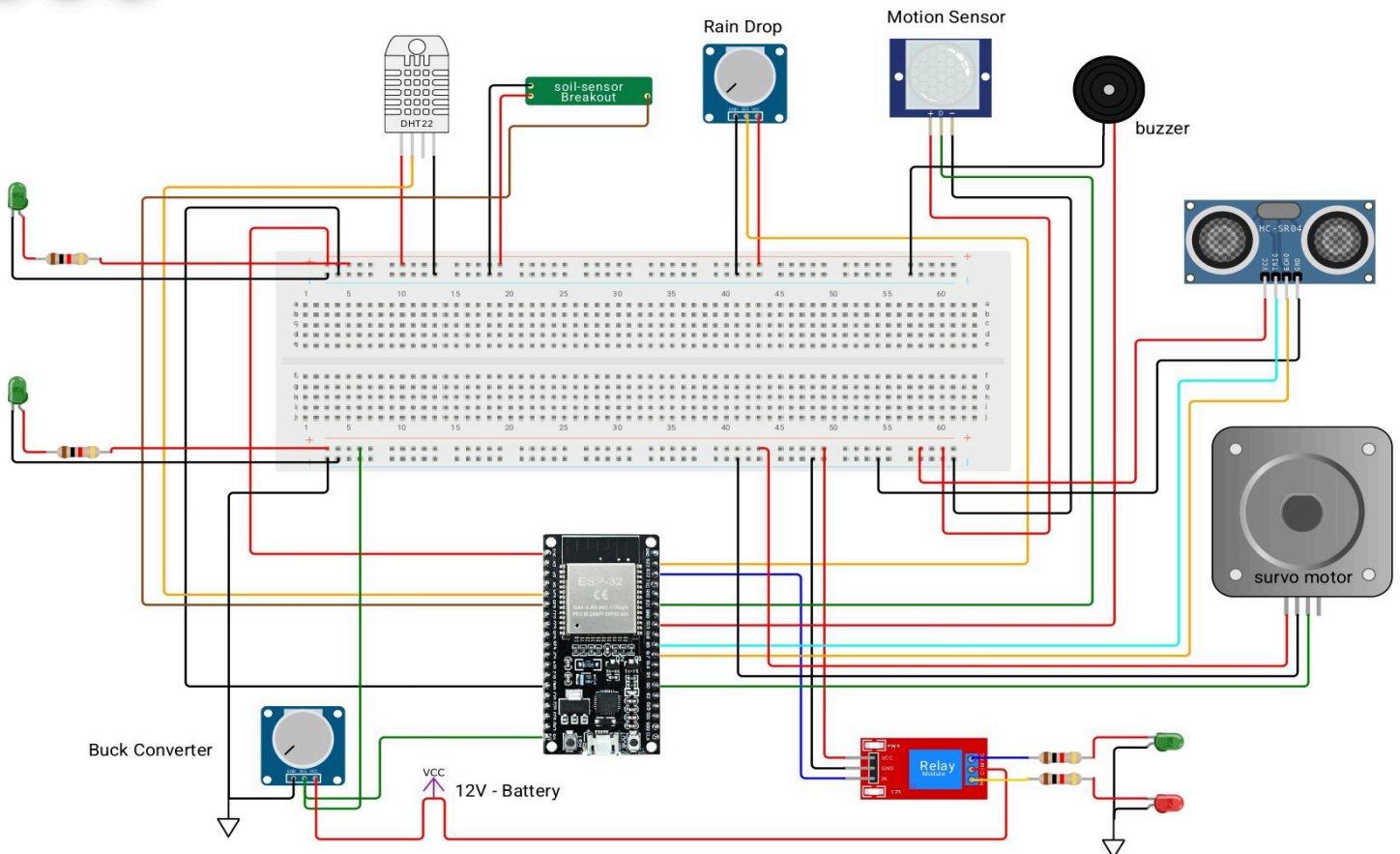
- **VS Code (Visual Studio Code):** The primary IDE for both full-stack web development and hardware firmware.
- **PlatformIO Extension:** Used for professional-grade ESP32 development, including library management and serial debugging.
- **Embedded C++:** The programming language used to write the logic for the ESP32.

8.3 Communication Protocols

- **HTTP/HTTPS (REST API):** Used by the ESP32 to POST sensor data to the Node.js backend.
- **WebSockets (Socket.io):** Enables real-time, bi-directional communication between the server and the React dashboard for instant pump control and live monitoring.

- **OpenWeatherMap API:** Integrated into the backend to fetch local weather data and optimize irrigation schedules.

9. Circuit Diagram



Circuit Description

The circuit represents a solar-powered smart agriculture system centered around an ESP32 microcontroller, which acts as the main control unit.

Environmental sensors such as the soil moisture sensor, DHT22 temperature and humidity sensor, raindrop sensor, and water level sensor continuously collect real-time

field and soil data. In addition to local sensor inputs, the system also utilizes external weather information obtained through a weather forecast API. Based on both sensor data and API-provided weather conditions, the ESP32 makes intelligent decisions to control irrigation and other farm operations.

A PIR motion sensor detects movement in the surrounding area while an ultrasonic sensor (HC-SR04) measures nearby distance to identify intrusion, and when motion or presence is detected, the ESP32 processes the signal and activates a buzzer to provide an audible alert for crop protection; additionally, a laser and LDR system monitors the farm perimeter using a continuous light beam, and if the beam is interrupted, it detects intrusion and triggers an alert.

The DC water pump is controlled using a relay module, allowing the low-voltage ESP32 to safely operate high-power irrigation equipment. A servo motor is included for automation tasks such as gate control or sensor adjustment.

Power is supplied by a 12V battery, regulated through a buck converter to provide stable voltage levels for the ESP32 and all connected sensors. LEDs are used as visual indicators to display system and relay status.

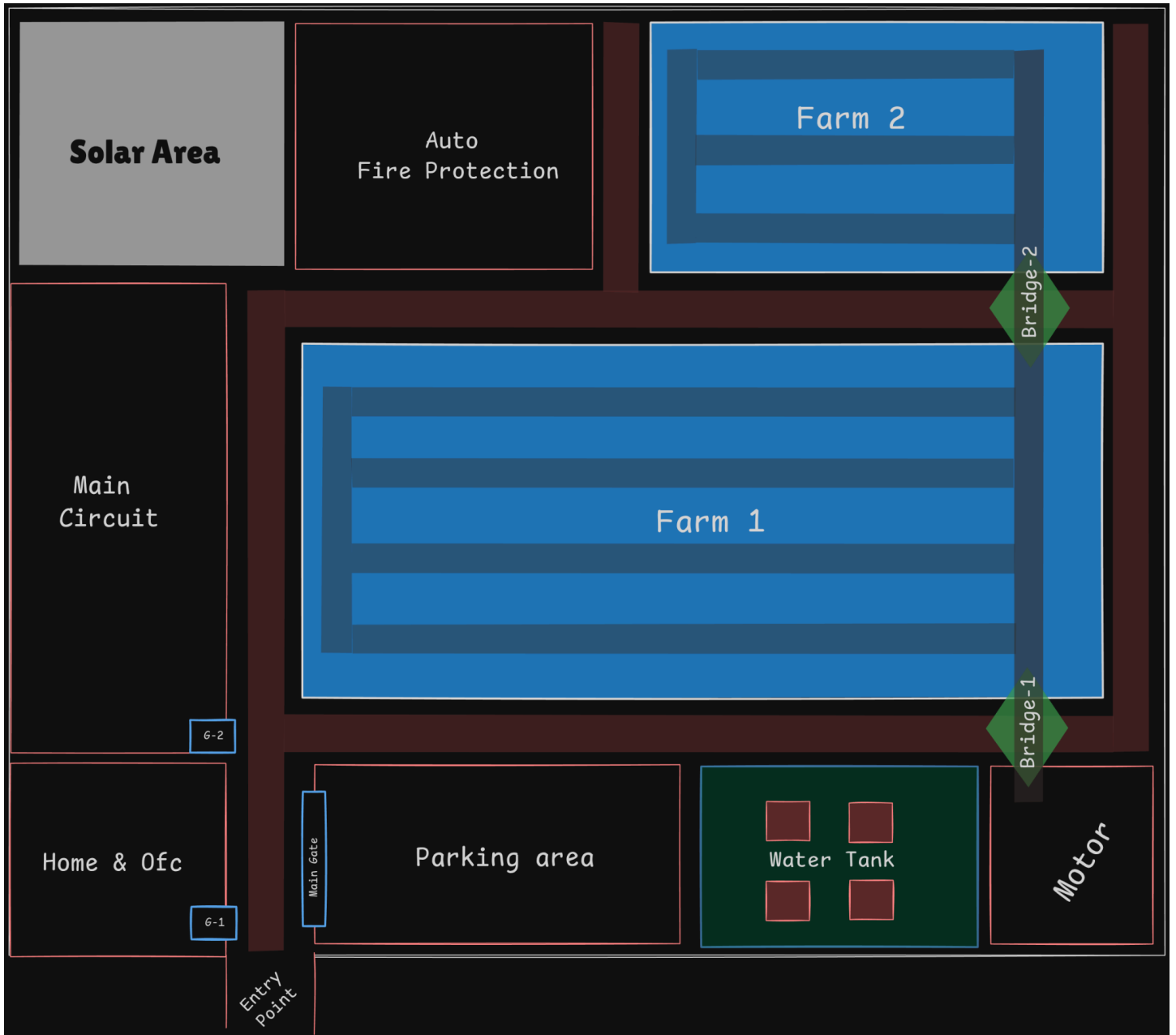
All collected sensor data, system responses, and operational status are transmitted to a web-based monitoring platform via APIs, where real-time information and overall system condition are displayed on the website. This allows users to remotely monitor farm conditions, irrigation status, and security alerts, enabling efficient and informed farm management.

10. Applications

The Digital Agro Farm system is designed for high versatility across several key sectors:

- **Precision Farming:** Automated irrigation for small-to-medium farms and greenhouses to optimize water and fertilizer use.
- **Off-Grid Agriculture:** Ideal for remote rural areas and rooftop gardens where grid electricity is unavailable or unstable.
- **Farm Security:** Real-time monitoring for animal intrusion or unauthorized movement using PIR and Ultrasonic sensors.
- **Academic Research:** A comprehensive prototype for studying IoT, Renewable Energy, and Data Science in agriculture.
- **Weather-Resilient Irrigation:** Smart pausing of water pumps during rainfall using local sensors and API integration.

11.BluePrint



12. Conclusion

Digital Agro Firm serves as a practical, scalable, and affordable solution for modernizing traditional agriculture. By seamlessly integrating Solar Energy, IoT-enabled Sensors, and Automated Irrigation, the system significantly reduces water waste, optimizes resource management, and enhances overall crop protection.

Unlike standard prototypes, this project demonstrates a sophisticated synergy between Hardware Engineering (ESP32) and Full-Stack Development (MERN). The implementation of a custom Node.js backend and React dashboard proves that low-cost technology, when paired with professional software architecture, can drive sustainable farming practices. Ultimately, this system provides a foundation for high-yield, data-driven agriculture that is both environmentally friendly and economically viable for the future.